

Saving America's Favorite Sport: By Saving its Players

Saving America's Favorite Sport: By Saving its Players

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Letter of Transmittal

Dear Professor Meierotto,

We wrote a magazine-style article titled "Saving America's Favorite Sport: By Saving its Players" which explores the impact of helmet design innovation and its role in player safety, focusing on concussions and chronic traumatic encephalopathy (CTE). This article analyzes how innovations in helmet design have reduced head injuries in football. Our intended audience includes football players, their families, and a general audience interested in sports safety and technology. The choice to do a magazine article allowed us to include technical content, while also making the information accessible to a wide audience.

Our research analyzes peer-reviewed studies on helmet designs and CTE as well as online sources that discuss the NFL's changes to helmet regulations and rules that promote safety. Our article begins by detailing background information about CTE. It then discusses early helmet design and some events that motivated improvements. Next, we talk about helmets that were standard in the 1990s, early 2000s, and current day innovations. Finally, we discuss why improvements to helmets alone are not sufficient for keeping players safe; other technology and rule/culture changes are needed to help reduce the likelihood of concussions and CTE.

The Hidden Cost of America's Favorite Sport

Every year, millions of Americans tune in to watch football players collide at full speed as helmets clash, crowds roar, and teams battle for victory. The sport has become a defining part of American culture, symbolizing athletic excellence, community pride, and national tradition. For players, football represents discipline, teamwork, and an opportunity to compete at the highest level. Yet, beneath the glory of the sport lies a growing awareness that the same collisions fueling football's spectacle may also endanger players long after the final whistle.

Chronic Traumatic Encephalopathy (CTE), a progressive neurodegenerative disease caused by repetitive head trauma, has become one of football's most troubling legacies. First identified in boxers in the 1920s, CTE has since been recognized in numerous National Football League (NFL) players, which has drawn significant attention to the long-term consequences of concussions and CTE in football (Lindsley, 2017). Some of these consequences include memory loss, confusion, impaired judgement, aggression, depression, and motor-skill difficulties (Lindsley, 2017). These symptoms reflect deeper structural changes within the brain itself. As shown in Figure 1, CTE is associated with a noticeable reduction in brain volume, which explains the cognitive, behavioral, and motor impairments observed in affected players. The very mechanisms that make football thrilling—the speed, strength, and force of collisions—also make it uniquely dangerous, particularly because of how the brain responds to impact forces.

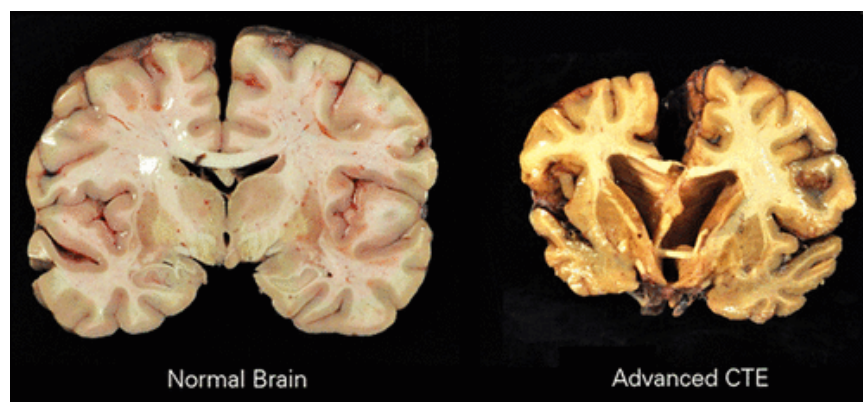


Figure 1. Comparison Between Healthy and CTE Brains. From “Chronic Traumatic Encephalopathy (CTE): A Brief Historical Overview and Recent Focus on NFL Players.” by Linsley, C., 2017, ACS Chemical Neuroscience, 8(8).

To understand why football players face such high injury risks, it helps to first look at what happens inside the brain during a hard hit. According to the Queensland Brain Institute, a forceful blow to the head can cause the brain to move either translationally or rotationally within the skull. In translational movement, which results from a direct impact, the brain ricochets back and forth against the sides of the skull, resulting in brain bruising. In rotational movement, the head is struck at an angle, which twists the brain and causes damage as neurons rub against each other (The University of Queensland, 2018). Both of these movements are common in football collisions and contribute to long-term brain injuries such as CTE.

What makes these forces so harmful is the delicate nature of the brain's structure. Unlike other organs, the brain is suspended in cerebrospinal fluid and encased by the rigid skull (Figure 2), making it vulnerable to sudden acceleration or deceleration. Even impacts that do not cause immediate symptoms, such as sub-concussive blows, can cause microscopic damage that accumulates over time. This is particularly concerning for offensive and defensive linemen, who

may experience dozens of small collisions in a single play. Over the course of a game, a season, or a career, these small, repetitive hits can become more dangerous than a single severe concussion.

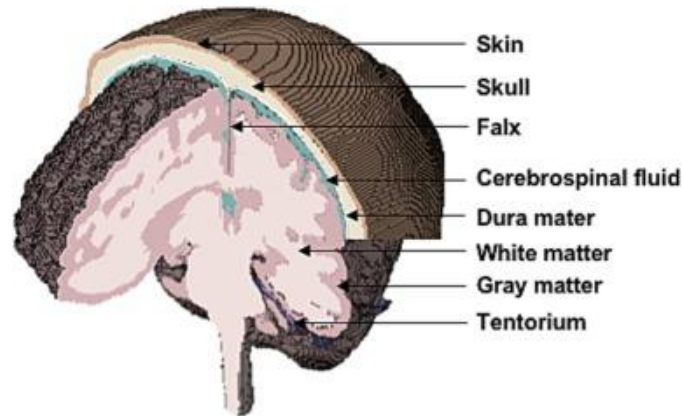


Figure 2. Layers of the Brain. From “Helmets with lattice liners can mitigate traumatic brain injury from impacts.” By Nasim, et al., 2024, *Journal of Biomechanics*, 176.

This understanding of how the brain is injured has become the foundation for efforts to make football safer. Over time, scientists, engineers, and football organizations have turned to helmet innovation as a way to reduce these brain-damaging forces and increase player safety. Modern helmets now reflect decades of research in materials science, biomechanics, and injury prevention. However, the question remains: are helmets enough, or is the sport's inherent physicality too dangerous to fully mitigate? As research continues to reveal the complexity of brain injury, helmet development has become not just an engineering challenge but a crucial part of determining how football can evolve while safeguarding the athletes who play it.

From Leather Caps to Plastic Shells: The Origins of Helmets

Football began picking up popularity as a sport in the late 1800s and early 1900s. Initially, players wore no head protection at all, or they wore simple hats made of soft leather and wool. Rather than protecting the skull and brain, this minimal equipment was designed to prevent abrasions and cauliflower ear. With few standardized rules, minimal officiating, and very little concern for player safety, early football was brutal and chaotic. Helmets were seen as optional accessories, not lifesaving equipment.

As the sport grew more competitive, its physicality intensified. Players and coaches exploited the lack of regulations, leading to a sharp rise in serious injuries, especially skull fractures and fatalities. Many of the most dangerous incidents came from head-to-head collisions or from formations like the infamous “flying wedge,” in which offensive players linked arms and charged forward as a human battering ram. The consequences were devastating. In 1905 alone, at least 18 players were killed (Abbott, 2011), a crisis that forced the sport to confront the urgent need for safer rules and equipment.

This football death crisis drew national scrutiny. President Theodore Roosevelt, concerned about the rising death toll, summoned football leaders to the White House and demanded reforms (Abbott, 2011). His intervention helped push for rule changes that outlawed dangerous formations and promoted safer styles of play. It also encouraged equipment manufacturers to focus more on preventing head injuries.

In response to Roosevelt's reforms, helmet makers began producing hardened leather helmets, which slightly improved protection (Stamp, 2012). Knowledge of brain injury mechanisms was limited at this time, but designers started focusing on impact absorption rather

than just skin protection. As displayed in Figure 3, these early leather helmets often included padded ridges or reinforced panels and were hand-stitched from multiple layers of leather. While they offered marginal protection from blunt impact, they were far from effective at reducing the forces associated with concussions.



Figure 3. An Early Football Cap. From “The Evolution of the Helmet.” By NFL, n.d.

As football continued to evolve, so did its headgear. The next catalyst for change was the increasing popularity of football following World War I and World War II. Soldiers returning from the war brought greater athletic discipline and a heightened interest in organized sports (Serb, 2019). Growing popularity pushed for more standardization and rudimentary safety testing. By 1943, helmets finally became mandatory in the NFL, and the first plastic helmets (depicted in Figure 4) became mainstream (MacDonald, 2020).



Figure 4. An Early Plastic Helmet. From “The Evolution of the Helmet.” By NFL, n.d.

These early plastic helmets marked a major shift in design philosophy. Inspired by military helmets, they were more durable, lighter, and better at retaining their shape than leather. The rigid polycarbonate shells provided improved impact distribution, reducing the likelihood of skull fractures. However, the padding inside was still quite basic, often consisting of simple foam or fabric liners (MacDonald, 2020).

As face injuries accumulated, the next innovations added to football helmets were air bladders and stainless-steel facemasks in the 1970s. The air bladders were an internal padding component that used air to cushion impacts, provide a secure fit, and improve protection. They allowed players to adjust the internal pressure for a more personalized fit, reducing excess head movement during collisions. Initially, facemasks were limited to a single plastic or rubber bar, but over time they evolved into stainless steel cages which offered players more facial protection from injuries like broken noses, gashes, and dental trauma (MacDonald, 2020).

As time continued to progress, helmet companies like Riddell continued to innovate. The iconic helmet of the 1990s (shown in Figure 5), called the VSR4, featured the first air-fitted liner

system. Unlike the fixed foam padding that was used before, the inflatable air liner provided a more snug fit, and better absorption of impact (Riddell, 2013). The VSR4 became one of the most widely used helmets in high school, college, and professional football, paving the way for the modern helmets used today.

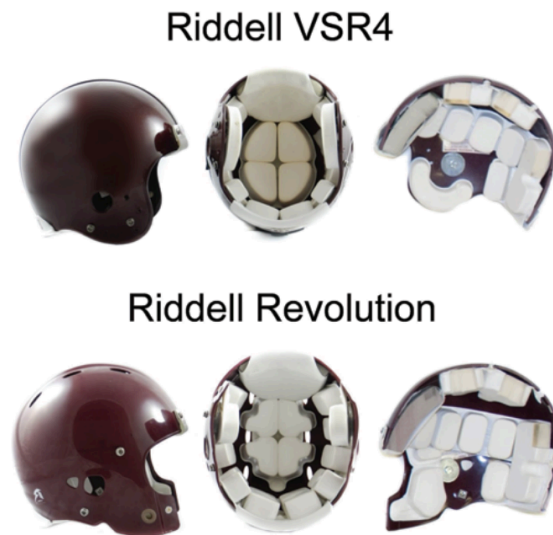


Figure 5. Riddell VSR4 Helmet. From “Lot Detail - 1990’s Tampa Bay Buccaneers Signed Full Size Riddell VSR4 Football Helmet.” By Mearsonlineauctions.com, 2015.

Modern Helmet Engineering: Advances, Innovations, and Testing

While the VSR4 was the standard throughout the 90s, it was replaced by the Riddell Revolution in 2002 as a result of the company’s continuing refinement (Riddell, 2013). As shown in Figure 6, the Revolution incorporated 40% more padding and a redesigned shell shape compared to the VSR4 (Rowson et al., 2014). These design features were engineered to address growing concerns about concussions by improving energy absorption and minimizing the movement of the head inside the helmet. The Revolution also included improved jaw padding

and a larger shell around the sides and back of the head, areas known to be vulnerable to concussive impacts (Ortman, 2012).



*Figure 6. Comparison Between Riddell VSR4 and Riddell Revolution. From “Can Helmet Design Reduce the Risk of Concussion in Football?” by Rowson et al., 2014, *Journal of Neurosurgery*, 120(4).*

To validate the Revolution's efficacy, a landmark 2014 study by Rowson et al. analyzed more than 1.2 million head impacts among collegiate players and found that the Revolution reduced concussion risk by roughly 50% compared with the older VSR4. Researchers attributed this improvement to the increased padding thickness and multi-layered foam engineered to dissipate impact forces more effectively (Rowson et al., 2014). Studies like this demonstrate that iterative engineering changes in helmet design can translate into measurable gains in player safety.

Additionally, a 2010 study by Casson et al. compared concussion rates in the NFL. This found that there were 0.38 documented concussions per NFL game in 2002-2007, representing a 7.6% reduction in concussion rate compared to the 1996-2001 seasons (Casson, 2010). This

lower concussion rate corresponded to the replacement of the VSR4 with the Revolution. However, the study also explained that in the 2002-2007 seasons, there was a worse return to play after head injuries (Casson, 2010). This demonstrated that the Revolution was far from a perfect solution. Even with substantial technological improvements, no helmet could prevent all head injuries.

In response to ongoing concussion concerns, the NFL continues to tighten performance requirements for all helmets. Each year, the NFL and NFL Players Association (NFLPA) test available models and publish safety rankings (NFL Football Operations, 2025). These tests simulate real-world impacts at various angles and speeds, emphasizing both linear and rotational acceleration. Helmets that do not meet standards are banned from use. Both the VSR4 and the Revolution have recently been added to the league's banned list in 2019 and 2025, respectively (NFL Player Health and Safety, 2025).

To accelerate innovation, the NFL recently launched a helmet-design competition, offering a \$3 million prize for the most protective helmet. The winning entries came from a new company named Light; its Apache Pro and Gladiator Thunder models were named among the league's Top-Performing helmets for the 2025 season (NFL Football Operations, 2025). Light's designs aim to improve player protection by dissipating more impact energy while being the most lightweight and comfortable helmet option (LIGHT Helmets, n.d.). These helmets, along with others in the NFL/NFLPA's Top-Performing group, seek to reduce concussion risk by minimizing the forces transmitted to the head during collisions (Zuckerman, 2019).

To effectively reduce these impact forces, modern helmets incorporate multiple layers of protection. Most designs begin with a rigid polycarbonate shell that distributes impact loads

across a larger surface area. Beneath the shell, manufacturers use multi-density foam layers tuned to absorb different impact energies. Many helmets also include inflatable liners and other fit-customization systems to ensure a secure, personalized fit. Newer models feature advanced, 3D-printed padding geometries engineered to optimize energy absorption. In addition, some helmets incorporate specialized internal or external structures that help reduce rotational acceleration, a key factor in brain injuries such as concussions and CTE (Nasim, et al., 2024).

However, even the newest helmets operate within the fundamental limits of what protective equipment can achieve. Their multilayered structures and optimized padding systems are designed to manage force as effectively as possible, but they cannot fully control the complex and unpredictable conditions of real play. As a result, evaluating helmet safety requires looking beyond design features alone and examining how these innovations perform when exposed to the actual demands of the sport.

Why Better Helmets Still Fall Short

Real-world data and biomechanical research show that current designs offer only partial protection against brain injury. Modern helmets have succeeded in reducing skull fractures and decreasing the severity of certain high-impact collisions, yet their actual impact on concussion prevention is far more modest. Large-scale studies, including those conducted in collegiate and professional football settings, consistently find that while newer helmets lower concussion risk, they do not eliminate it. This gap between technological improvement and clinical outcome highlights the constraints that even the best helmets face. Helmets are designed primarily to absorb linear forces and protect against skull fractures, but they cannot completely eliminate the rotational acceleration that causes the brain to move within the skull (Yengo-Kahn et al., 2016).

Because rotational forces play a central role in the biomechanics of concussion, this limitation remains one of the biggest engineering challenges in helmet design.

An additional drawback is that helmets are most effective in controlled laboratory tests, where conditions can be standardized, impacts can be precisely replicated, and forces are measured in isolation. Actual football play is far more chaotic. Impacts vary widely in direction, timing, point of contact, and the athlete's preparedness at the moment of collision. The variability of these real-world conditions means that even top-performing helmets cannot consistently deliver the same level of protection observed in test environments. This discrepancy underscores the challenge of engineering a single piece of equipment that must perform reliably across thousands of unpredictable scenarios.

Another significant limitation of helmet-based injury prevention involves the structure of football itself. The majority of head impacts in football are not isolated blows but part of repetitive contact patterns embedded in normal gameplay. Linemen, for example, experience frequent low-to-moderate impacts on nearly every snap, many of which occur too quickly or subtly to activate the helmet's protective features in a meaningful way. These accumulated exposures, rather than a single dramatic hit, pose the greatest long-term neurological risks. Because helmets are designed to manage force at the moment of impact, not to counteract cumulative microtrauma, they are inherently limited in addressing the forms of injury now understood to contribute most heavily to CTE.

Resource disparities also affect the impact of helmet technology. Professional and well-funded college programs have access to the newest models, routine equipment replacement, and advanced fitting systems. In contrast, many youth and high school programs rely on older

helmets that may not meet current safety standards. Reconditioning programs can extend helmet lifespan, but they cannot fully compensate for outdated designs or worn materials. As a result, improvements in helmet engineering do not benefit all athletes equally, and concussion risk remains uneven across different levels of play.

Behavioral and cultural factors further constrain helmet effectiveness. Football remains a sport built on aggressive contact, competitive intensity, and a strong emphasis on toughness. Even as awareness of brain injury rises, players may downplay symptoms, rush back into play, or adopt hazardous tackling techniques. Coaches may unintentionally reinforce risky behaviors, and athletes at all levels often associate protective equipment with invulnerability. This combination of cultural pressure and risk compensation can offset the benefits offered by advanced helmets.

Lastly, improved concussion statistics must be interpreted cautiously. Apparent decreases in injury rates may reflect changes in reporting practices, increased medical oversight, or stricter protocol enforcement rather than direct effects of helmet design. Conversely, higher concussion numbers in certain seasons may indicate better recognition rather than worsening safety. These confounding factors make it difficult to attribute epidemiological trends solely to equipment improvements.

Taken together, these limitations demonstrate that helmet technology, while essential, cannot fully solve the complex problem of brain injury in football. Helmets protect the skull and reduce certain forces, but they cannot prevent many of the conditions that produce concussions or long-term neurodegeneration.

Beyond the Helmet: Rule Changes, Training, and Medical Protocols

As Yengo-Kahn et al. (2016) emphasize, helmet innovation alone cannot eliminate concussions in football; it must be paired with ongoing education, rule enforcement, and cultural changes that prioritize player health.

One way to do this is through coaching and training practice. Teaching athletes to avoid leading with the head, reinforcing proper tackling techniques, and structuring practices with fewer full-contact drills can significantly reduce high-risk impacts. Youth leagues, high schools, and collegiate programs have increasingly adopted “heads-up” tackling instruction and limited-contact practice schedules. As Schussler et al. (2018) indicated, implementing these “heads up” tackling techniques decreased the number of dangerous head accelerations.

Rule enforcement and game regulation are equally important. Football leagues have introduced targeting penalties, kick-off rule adjustments, and stricter enforcement of roughing-the-passer rules to remove some of the most dangerous collision scenarios from the game. Importantly, data shows that these measures are effective. A national surveillance study of high-school football players found that after the 2014 rule limiting helmet-to-helmet contact, the rate of emergency-department diagnosed concussions declined by 6.2% between 2015 and 2019, compared with a 10.7% increase between 2009 and 2013 (Obana et al., 2021). These findings demonstrate that changes in rule enforcement can meaningfully reduce in-game injury rates and complement the protective role of helmets.

Another indispensable pillar of protection is medical evaluation and monitoring. Sideline neurologists, independent concussion spotters, and standardized return-to-play protocols provide more reliable oversight than the informal evaluations common in past decades. Baseline

cognitive testing and post-impact assessments allow for earlier detection of injuries that might otherwise go unnoticed. In addition, impact-monitoring sensors, which are now implemented in many advanced helmets, provide real-time data about collision frequency and severity. These tools give medical teams improved ability to track player exposure throughout a season (Around the NFL Staff, 2023).

Finally, continued scientific research remains essential. The biomechanics of brain injury, particularly the long-term effects of repetitive sub-concussive trauma, are still being understood. Ongoing studies in neuroimaging, material science, and computational modeling will be crucial for guiding future rule changes and informing the next generation of protective equipment.

Taken together, these non-equipment-based interventions show that real progress in player safety requires cooperation across engineering, coaching, policy making, and medicine. Helmets can reduce risk, but the responsibility for protecting athletes must be shared across every level of the sport. This broader perspective provides the foundation for understanding what the future of football safety must look like.

The Future of Football Safety: Engineering, Medicine, and the Path Forward

As football continues to evolve, one truth has become clear: the future of player safety will depend not only on how the game is played, but also on how effectively science and engineering can continue to push the boundaries of helmet technology. Modern helmets have come a long way from the soft leather caps of the early 1900s, yet current designs, no matter how advanced, remain constrained by the basic biomechanics of the brain. The next generation of helmets must confront challenges that previous technologies could not fully address, particularly the reduction of rotational forces and the cumulative effects of sub-concussive impacts.

Future helmet research is moving toward adaptive, data-informed systems that can respond dynamically to impacts rather than relying solely on passive padding. Engineers are exploring 3D-printed lattice structures, shear-thickening materials, and sliding-shell mechanisms that can better manage rotational acceleration. Nasim, et al. (2024) analyzed how a polyamide 12 material can be rearranged in different lattice structures to provide better protection against rotational impacts to the brain. These concepts represent more than incremental upgrades; they offer a chance to fundamentally rethink how helmets manage force.

Perhaps most importantly, future research must focus on personalized helmet design. Advances in 3D scanning and custom-fitting methods, already seen in helmets like the Riddell Axiom, suggest that individualized protection may become standard (Riddell | Axiom, n.d.). By tailoring the internal padding, shell geometry, and impact response to each athlete's head shape and playing style, engineers can optimize safety in ways that universal designs cannot.

Still, even the most revolutionary helmet technologies will not entirely eliminate brain injury from football. The brain's vulnerability to rapid acceleration is a biological reality that equipment alone cannot fully overcome. But through sustained investment in scientific research, innovative materials, computational modeling, and player monitoring systems, we can continue to shrink the gap between what helmets currently do and what athletes truly need.

The future of helmet research is not simply about making better equipment; it is about redefining what it means for football to be safe. If researchers, engineers, policymakers, and football organizations work together, the sport can continue to thrive while providing players with a safer environment. In this way, advancing helmet technology is not just about innovation; it is about preserving the health of athletes and securing the future of America's favorite sport.

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